Transkriptomanalyse

11. November 2026, Sommersemester 2025

Aufgabe 1

a)

Es werden die Definitionen von g_n, f_n, $\bar{b}_n$, $\bar{c}_n$, $\bar{y}_n$ und $\bar{z}_n$ als gegeben (wie im Vorlesungsskript) vorausgesetzt.

$$
\begin{aligned}
g_n &= \log_2(f_n) \\
&= \log\left(\frac{\bar{b}_n}{\bar{c}_n}\right) \\
&= \log(\bar{b}_n) - \log(\bar{c}_n) \\
&= \log\left(\sqrt[L_1]{\prod_{l=1}^{L_1} b_{n,l}}\right) - \log\left(\sqrt[L_2]{\prod_{l=1}^{L_2} c_{n,l}}\right) \\
&= \log\left(\left(\prod_{l=1}^{L_1} b_{n,l}\right)^{\frac{1}{L_1}}\right) - \log\left(\left(\prod_{l=1}^{L_2} c_{n,l}\right)^{\frac{1}{L_2}}\right) \\
&= \frac{1}{L_1} \cdot \log\left(\prod_{l=1}^{L_1} b_{n,l}\right) - \frac{1}{L_2} \cdot \log\left(\prod_{l=1}^{L_2} c_{n,l}\right) \\
&= \frac{1}{L_1} \cdot (\log(b_{n,1} \cdot b_{n,2} \cdot \cdots \cdot b_{n,L_1})) - \frac{1}{L_2} \cdot (\log(c_{n,1} \cdot c_{n,2} \cdot \cdots \cdot c_{n,L_2})) \\
&= \frac{1}{L_1} \cdot (\log(b_{n,1}) + \log(b_{n,2}) + \cdots + \log(b_{n,L_1})) - \frac{1}{L_2} \cdot (\log(c_{n,1}) + \log(c_{n,2}) + \cdots + \log(c_{n,L_2})) \\
&= \frac{1}{L_1} \cdot \left(\sum_{l=1}^{L_1} \log(b_{n,l})\right) - \frac{1}{L_2} \cdot \left(\sum_{l=1}^{L_2} \log(c_{n,l})\right) \\
&= \frac{1}{L_1} \cdot \sum_{l=1}^{L_1} y_{n,l} - \frac{1}{L_2} \cdot \sum_{l=1}^{L_2} z_{n,l} \\
&= \bar{y}_n - \bar{z}_n
\end{aligned}
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